Ahmed FAKHFAKH

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AI Engineer | Data Engineer | MLOps | Edge AI

Seeking a Permanent Position – Available Immediately

Data and AI Engineer from Polytech Nantes (2026) with 3 years of embedded AI R&D apprenticeship at OPmobility. I productionize AI systems, from GCP data pipelines to optimized inference: multi-agent LLM and RAG, MLOps, Deep Learning on NVIDIA Jetson (TensorRT). Google Cloud Certified Professional Data Engineer.

CERTIFICATIONS

Google Cloud Certified – Professional Data Engineer | Associate Cloud Engineer | Cloud Digital Leader

SKILLS

- **Languages & Cloud:** Python, SQL, Bash, R, GCP (BigQuery, Vertex AI, Cloud Run, Cloud Build, Cloud Storage)
- **AI & LLM:** PyTorch, Scikit-Learn, XGBoost, LangGraph, LlamaIndex, Qdrant, Hugging Face (LoRA)
- **Edge AI & Optimization:** TensorRT (FP16/INT8), ONNX, NVIDIA Jetson, GPU profiling (Perfetto), sparse models
- **Data Engineering & MLOps:** Apache Airflow, dbt, Spark, Kafka, Flink, Iceberg, MLflow, DVC, LangSmith
- **DevOps & Tools:** Terraform, Docker, CI/CD (Cloud Build, GitHub Actions, GitLab CI), FastAPI, Git, Linux

WORK EXPERIENCE

Embedded AI Engineer (Apprenticeship) | OPmobility (OP'nSoft) | Levallois-Perret **Oct. 2023 – Sep. 2026**

- **Edge AI Deployment & Profiling:** Ported a 4D radar detection model to an NVIDIA Jetson AGX Orin (Hardware-in-the-Loop bench). TensorRT FP16: GPU latency from 865 to 188 ms (4.6x), RAM from 10.2 to 3.5 GB, accuracy preserved (Pd 0.58 vs 0.59).
- **Optimization & Diagnosis:** Ablation study (sparse networks, MobileNet-V2, FIFO cache with multiprocessing, distillation, INT8); diagnosed the *mixed-precision fallback* that made INT8 slower than FP16.
- **Attention R&D (co-author, IEEE RadarConf 2027):** Radar tokenizer and asymmetric-attention transformer: same detection as the CNN baseline with 3.6x fewer operations, 21 FPS on Jetson Orin (vs 7.2), 4x less energy.
- **Technical Stack:** Python, PyTorch, TensorRT, ONNX, CUDA, NVIDIA Jetson, Multiprocessing.

PUBLICATION

Tailored Attention Mechanism for Radar Point-Cloud Generation on Embedded GPUs – A. Hernandez, A. Pegatoquet, J.-Y. Dauvignac, R. Christy-Krishanth, **A. Fakhfakh**. *Submitted to IEEE RadarConf 2027.*

TECHNICAL PROJECTS

Multimodal Financial Agentic AI – Multi-Agent Architecture & LLMOps GitHub Link | **2025 – 2026**

- LangGraph supervisor routing each request to 4 agents: self-correcting BigQuery SQL, document RAG (ChromaDB), real-time stock prices, conversational.
- Anti prompt-injection guardrail at the graph entry point; LangSmith tracing of latency and cost per agent (about $0.007 per request).
- Serverless Docker deployment on Cloud Run (streaming Streamlit UI), Terraform infrastructure.
- **Stack & Tools:** Vertex AI Gemini, LangGraph, LangChain, BigQuery, ChromaDB, LangSmith, Terraform.

Modern Data Stack & RAG with Hallucination Detection GitHub Link | **2025 – 2026**

- ELT pipeline: IT support ticket cleaning with dbt on BigQuery (integrity tests), Vertex AI embeddings and Qdrant indexing.
- LlamaIndex RAG on Cloud Run, with automatic evaluation of every answer (faithfulness, relevancy) and source display.
- Daily serverless orchestration (Cloud Scheduler + Cloud Run Jobs), cheaper than Airflow.
- **Stack & Tools:** dbt, BigQuery, Vertex AI, LlamaIndex, Qdrant, Cloud Run, Docker.

Other projects: Fraud detection streaming (Kafka, PyFlink, Iceberg) | QLoRA fine-tuning of Llama-3 8B (MLflow, DVC) | XGBoost MLOps API (FastAPI, Cloud Build, Vertex AI).

EDUCATION

Engineering Degree in Data & AI (Master's level) | Polytech Nantes, France **Oct. 2023 - Sep. 2026**
Courses: Data Science, Machine Learning, Deep Learning, Computer Vision, MLOps.

DUT in Computer Science (two-year degree) | IUT de Bayonne, France **Sep. 2021 - Jun. 2023**
Courses: Mathematical Analysis, Algorithms, Object-Oriented Programming, SQL.

LANGUAGES

English TOEIC 830 | **French** Fluent | **Arabic** Native